8. Measure fluorescence of sample using instrument parameters that correspond to those used when generating standard curve (see steps 2 to 6). To minimize photobleaching effects, keep time for fluorescence measurement constant for all samples.

9. Subtract fluorescence value of reagent blank from that of each sample. Determine DNA concentration of the sample from standard curve.

10. If desired, repeat assay using a different dilution of the sample to confirm results.

QUANTITATION OF NUCLEIC ACIDS USING A MICROVOLUME SPECTROPHOTOMETER WITHOUT THE USE OF CUVETTES OR CAPILLARIES

ALTERNATE PROTOCOL 4

The NanoDrop ND-1000 spectrophotometer uses a patented sample retention system (Fig. A.3D.2) that holds 1 µl of sample without the need for traditional containment devices such as cuvettes and capillaries. Using fiber optic technology and surface tension, the sample is held in place between two optical surfaces that define the pathlength in a vertical orientation. Removal of fixed containment devices from the system allows the pathlength to change in real time for a given sample. During each measurement cycle, the sample is assessed at both a 1-mm and 0.2-mm path, providing an extensive dynamic range (2 ng/µl to 3700 ng/µl for dsDNA; Table A.3D.5). This essentially eliminates the need to perform dilutions or to make any assumptions regarding the sample concentration prior to the measurement. Direct coupling of the sample to the optics of the spectrophotometer removes interference caused by incident light and transmitted light passing through containment walls of traditional cuvettes, microcell cuvettes, and capillaries. Preparation for the next sample only requires wiping of both optical surfaces with a common laboratory wipe. The time needed for performing dilutions and for cleaning the cuvette and microcell devices, as well as the possibility of damaging such devices, is eliminated. Total measurement cycle time, including preparation and removal of the sample, is ~30 sec. The ease of use of this protocol not only makes it a feasible option for small volume analysis, but also a practical alternative for all spectrophotometric measurements.

Materials

Nucleic acid sample to be quantitated
Water or buffer in which sample is dissolved
NanoDrop ND-1000 Spectrophotometer (NanoDrop Technologies, Inc., *http://www.nanodrop.com*)

1. Clean the upper and lower optical surfaces of the microspectrophotometer sample retention system as follows. Pipet 1 to 2 µl of clean deionized water onto the lower optical surface. Close the lever arm and tap it a few times to bathe the upper optical surface. Lift the lever arm and wipe off both optical surfaces with a Kimwipe.

   Wiping the pedestals between measurements is usually sufficient to prevent sample carryover and avoid residue buildup. A final cleaning of all surfaces with deionized water is recommended after each user's last measurement. After a large number of samples, the areas around the pedestals should be cleaned thoroughly to prevent previous samples from being wiped back onto the pedestals, affecting low-level measurements. It may be advisable to clean the measurement surfaces with 2 µl water (as above) after particularly high-concentration samples.

   The NanoDrop ND-1000 spectrophotometer is compatible with most solvents typically used in life science laboratories, including methanol, ethanol, n-propanol, isopropanol, butanol, acetone, ether, chloroform, carbon tetrachloride, DMSO, DMF, acetonitrile, THF, toluene, hexane, benzene, sodium hydroxide, sodium hypochlorite (bleach), dilute HCl, dilute HNO_3, and dilute acetic acid.

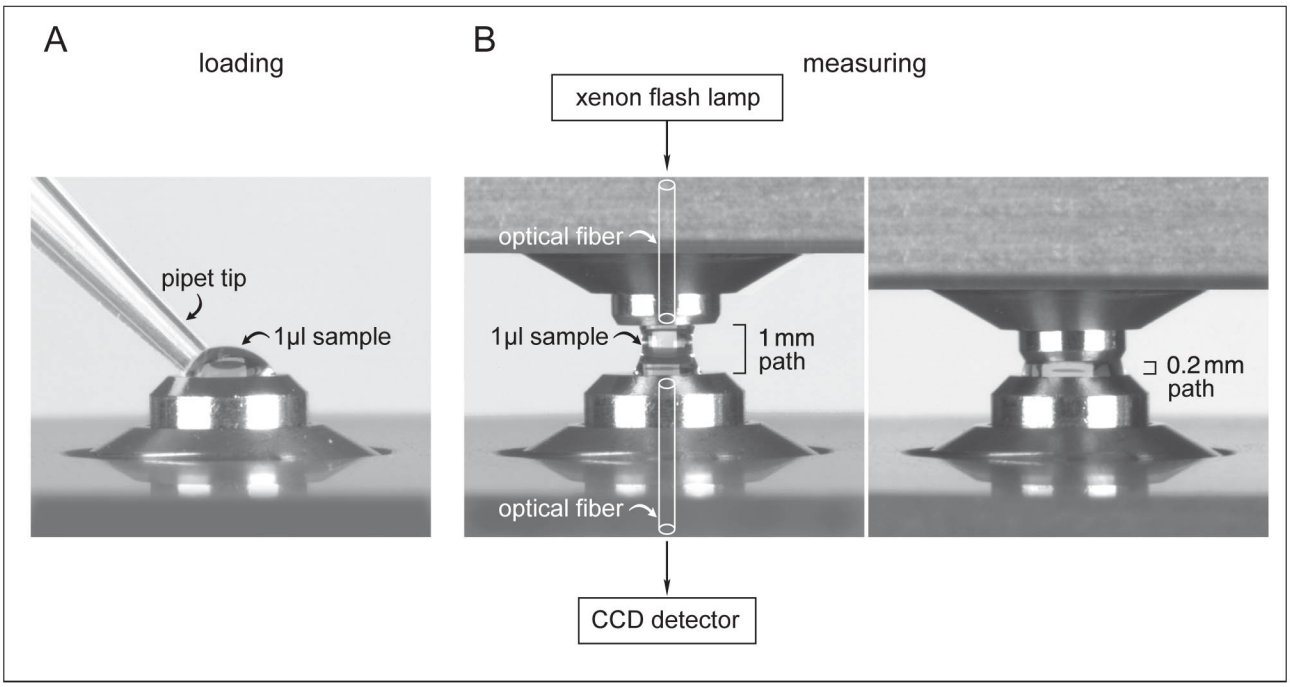

Figure A.3D.2 The NanoDrop ND-1000 Spectrophotometer microvolume sample retention system. (**A**) A sample volume of 1 µl is dispensed onto the lower optical surface. (**B**) Once the instrument lever arm is lowered, the upper optical surface engages with the sample, forming a liquid column with the path length defined by the gap between the two optical surfaces. During each measurement, the sample is assessed at both a 1-mm and 0.2-mm path, providing a large dynamic range of nucleic acid detection.

CAUTION: *Hydrofluoric acid (HF) must not be used to clean the NanoDrop instrument, as the fluoride ion will dissolve the quartz fiber optic cable.*

2. Open the NanoDrop software and select the nucleic acids module.

3. Initialize the spectrophotometer by placing 1 µl clean water onto the lower optic surface, lowering the lever arm, and selecting "initialize" in the NanoDrop software. Once initialization is complete (~10 sec), clean both optical surfaces with a Kimwipe.

4. Perform a blank measurement by loading 1 µl deionized water or buffer and selecting "blank." Once the blank is complete, clean both optical surfaces with a Kimwipe.

   As on a traditional spectrophotometer, the blank will be subtracted from subsequent measurements. To determine the contribution of a blank, such as a specific buffer, measure the buffer against deionized water as the blank. If the buffer does not contribute to the A_{260}, then deionized water will suffice as the blank. The water or buffer should always be measured to ensure that the instrument has been zeroed properly. The measurement of the water or buffer should be zero or very close to zero.

   All measurements are automatically normalized to 340 nm.

5. Measure the nucleic acid sample by loading 1 µl and selecting "measure." Once the measurement is complete, clean both optical surfaces with a Kimwipe.

   The measurement and subsequent concentration calculation is volume independent. The sample only needs to bridge the gap between the two optical surfaces for a measurement to be made. Field experience indicates that a 1-µl volume is sufficient to ensure reproducibility of aqueous solutions of nucleic acids. Volumes of 2 µl may be used for samples suspected of containing substances that may affect surface tension.

   Ensure that the appropriate constant has been chosen for the measurement (e.g., 50 for dsDNA or 40 for RNA). The software automatically calculates the nucleic acid concentration. If the calculation is being done by hand, the A_{260} is represented as a 1-cm path for convenience (even though 1-mm and 0.2-mm paths are actually used during the measurement cycle).